Chapter 9. Factor Analysis

Model, estimation (principal components, principal factors, MLE), rotation, factor scores

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1 The Orthogonal Factor Model

Motivation. Factor analysis explains correlations by a few unobserved common factors: $\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi}$. Unlike PCA, it is a model with specific error variances, and the loadings are determined only up to rotation.

The covariance of p observed variables is explained by a few, k, unobserved common factors.

Definition 9.1: factor model

$$X_i = \mu_i + \sum_{\ell=1}^{k} \lambda_{i\ell} F_\ell + U_i \quad (i = 1, \dots, p), \qquad \mathbf{X} = \boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{F} + \mathbf{U},$$

where $\boldsymbol{\Lambda} = (\lambda_{i\ell})$ is the $p \times k$ **loading matrix**, $\mathbf{F} \in \mathbb{R}^k$ the common factors and $\mathbf{U} \in \mathbb{R}^p$ the specific factors, with

$$\mathrm{E}(\mathbf{F}) = \mathbf{0}, \quad \mathrm{Cov}(\mathbf{F}) = \mathbf{I}_k, \quad \mathrm{E}(\mathbf{U}) = \mathbf{0}, \quad \mathrm{Cov}(\mathbf{U}) = \boldsymbol{\Psi} = \mathrm{diag}(\psi_1, \dots, \psi_p), \quad \mathrm{Cov}(\mathbf{F}, \mathbf{U}) = \mathbf{0}.$$

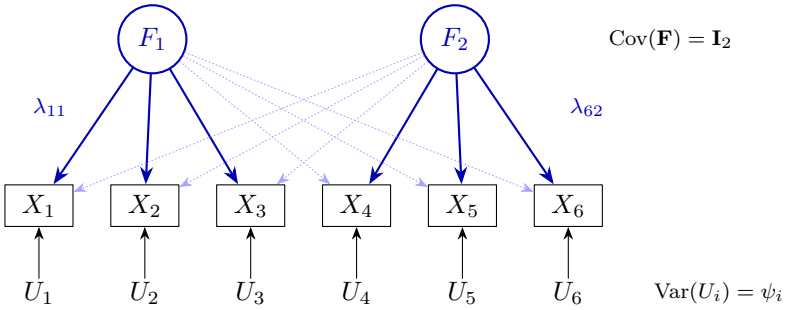


Figure. *Path diagram of a two-factor model for six variables. Every factor may load on every variable (dotted arrows carry small loadings in a simple structure), each variable has its own specific factor U_i, and the U_i are uncorrelated with each other and with $\mathbf{F}$. So any correlation between two X's must pass through the common factors:* $\mathrm{Cov}(X_i, X_j) = \sum_\ell \lambda_{i\ell}\lambda_{j\ell}$.

Theorem 9.2: covariance structure

$$\mathrm{E}(\mathbf{X}) = \boldsymbol{\mu}, \qquad \mathrm{Cov}(\mathbf{X}) = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}, \qquad \mathrm{Cov}(\mathbf{X}, \mathbf{F}) = \boldsymbol{\Lambda},$$

$$\mathrm{Var}(X_i) = \underbrace{\sum_{\ell=1}^{k} \lambda_{i\ell}^2}_{h_i^2\text{: communality}} + \underbrace{\psi_i}_{\text{specific variance}}, \qquad \mathrm{Cov}(X_i, X_j) = \sum_\ell \lambda_{i\ell}\lambda_{j\ell} \ (i \neq j).$$

Proof. $\mathrm{Cov}(\boldsymbol{\Lambda}\mathbf{F} + \mathbf{U}) = \boldsymbol{\Lambda}\mathrm{Cov}(\mathbf{F})\boldsymbol{\Lambda}^\top + \mathrm{Cov}(\mathbf{U}) + \boldsymbol{\Lambda}\mathrm{Cov}(\mathbf{F}, \mathbf{U}) + \mathrm{Cov}(\mathbf{U}, \mathbf{F})\boldsymbol{\Lambda}^\top = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}$, and $\mathrm{Cov}(\mathbf{X}, \mathbf{F}) = \mathrm{Cov}(\boldsymbol{\Lambda}\mathbf{F} + \mathbf{U}, \mathbf{F}) = \boldsymbol{\Lambda}$. ■

Because $\boldsymbol{\Psi}$ is diagonal, covariances between variables come *entirely* from the common factors. The model explains the $p(p+1)/2$ entries of $\boldsymbol{\Sigma}$ with $pk + p$ parameters.

1.1 Non-identifiability and rotation

For orthogonal $\mathbf{T}$ $(k \times k)$, $\boldsymbol{\Lambda}^* = \boldsymbol{\Lambda}\mathbf{T}$ and $\mathbf{F}^* = \mathbf{T}^\top\mathbf{F}$ give $\boldsymbol{\Lambda}^*\mathbf{F}^* = \boldsymbol{\Lambda}\mathbf{F}$, $\mathrm{Cov}\mathbf{F}^* = \mathbf{I}_k$ and $\boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top$: loadings are determined only up to rotation. If $\hat{\boldsymbol{\Lambda}}$ is an MLE, so is $\hat{\boldsymbol{\Lambda}}\mathbf{R}$ for any orthogonal $\mathbf{R}$. Removing the $k(k-1)/2$ rotational degrees of freedom leaves $pk + p - k(k-1)/2$ parameters

and

$$\nu = \frac{p(p+1)}{2} - \left\{pk + p - \frac{k(k-1)}{2}\right\} = \frac{(p-k)^2 - p - k}{2},$$

which must be positive for the model to impose a real restriction.

1.2 Invariance to scale changes

If $Y_i = c_i X_i$, i.e. $\mathbf{Y} = \mathrm{diag}(\mathbf{c})\mathbf{X}$, then $\mathbf{Y} = \mathrm{diag}(\mathbf{c})\boldsymbol{\mu} + \mathrm{diag}(\mathbf{c})\boldsymbol{\Lambda}\mathbf{F} + \mathrm{diag}(\mathbf{c})\mathbf{U}$ is again a factor model with $\boldsymbol{\Lambda}_Y = \mathrm{diag}(\mathbf{c})\boldsymbol{\Lambda}$ and $\boldsymbol{\Psi}_Y = \mathrm{diag}(\mathbf{c})^2\boldsymbol{\Psi}$. For standardized variables ($\mathrm{Var} X_i = 1$),

$$\lambda_{i\ell} = \mathrm{Cov}(X_i, F_\ell) = \mathrm{Corr}(X_i, F_\ell), \qquad \mathrm{Corr}(Y_i, F_\ell) = \lambda_{i\ell},$$

so loadings are read as variable–factor correlations. In practice factor analysis is often run on the correlation matrix $\mathbf{R}$.

2 Estimation

2.1 Principal component method

From $\widehat{\mathrm{Corr}}(\mathbf{X}) = \mathbf{R} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2} = \sum_{j=1}^{p} l_j \mathbf{u}_j \mathbf{u}_j^\top \approx \sum_{j=1}^{k} l_j \mathbf{u}_j \mathbf{u}_j^\top$ (largest k eigenvalues),

$$\hat{\boldsymbol{\Lambda}} = [\sqrt{l_1}\mathbf{u}_1, \ldots, \sqrt{l_k}\mathbf{u}_k], \qquad \hat{\boldsymbol{\Psi}} = \mathrm{diag}(\mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top).$$

The diagonal of $\mathbf{R}$ is reproduced exactly, and the sum of squared off-diagonal residuals of $\mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top - \hat{\boldsymbol{\Psi}}$ is at most $l_{k+1}^2 + \cdots + l_p^2$. Factor ℓ explains a proportion l_ℓ/p of the total variance. Since eigenvalues do not change, increasing k leaves earlier loadings unchanged.

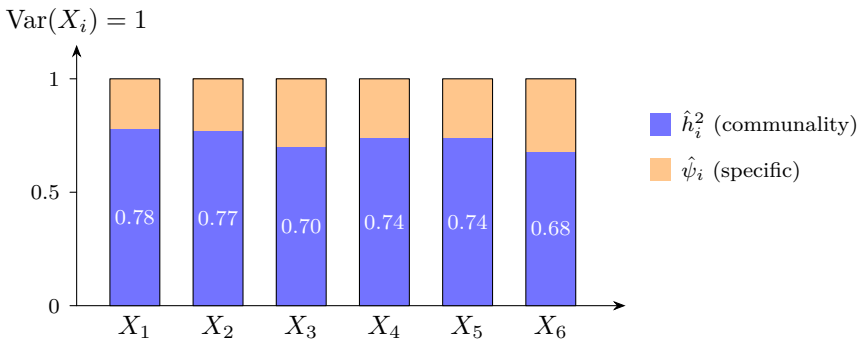


Figure. *Principal component solution with $k = 2$ for a synthetic correlation matrix ($n = 300$, $p = 6$; $l_1 = 2.43$, $l_2 = 1.96$, $l_3 = 0.48$). Each unit bar is $1 = \hat{h}_i^2 + \hat{\psi}_i$: the diagonal of $\mathbf{R}$ is reproduced exactly, and the two factors explain $(l_1 + l_2)/p = 73\%$ of the total standardized variance.*

Remark 9.3: number of factors with the principal component method

Heuristics are used when PCA is used to estimate the factor loadings:

1. use a **scree plot** or a **cumulative scree plot** of $l_1 \geq l_2 \geq \cdots \geq l_p$;
2. or use **Kaiser's rule**: retain factors $1, \ldots, k$ satisfying

$$l_k > \bar{l} = \frac{1}{p}\sum_{i=1}^{p} l_i.$$

When correlation PCA is used, $\sum_i l_i = \mathrm{tr}\mathbf{R} = p$, so $\bar{l} = 1$ and the rule keeps factors with $l_k > 1$ (a factor must explain more than one standardized variable's worth of variance).

These are the same rules as for choosing the number of principal components (Ch. 7); the likelihood ratio test of §2.4 is available only under the normal MLE approach.

2.2 Principal factor method

Starting from initial communality estimates $\hat{h}_i^2$ (e.g. $1 - 1/r^{ii}$, r^{ii} the diagonal of $\mathbf{R}^{-1}$), form the reduced correlation matrix $\mathbf{R} - \hat{\boldsymbol{\Psi}}$ and iterate

1. $\mathbf{R} - \hat{\boldsymbol{\Psi}} \approx \hat{\boldsymbol{\Lambda}}_{(k)} \hat{\boldsymbol{\Lambda}}_{(k)}^\top$ (top-k eigendecomposition of the reduced matrix);
2. $\hat{\psi}_{(k)i} = 1 - \sum_{\ell=1}^{k} \hat{\lambda}_{(k)i\ell}^2$

until convergence.

2.3 Maximum likelihood

Assume $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})$. Profiling $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$, twice the negative log-likelihood is, up to constants and a factor n,

$$\ell(\hat{\boldsymbol{\mu}}, \boldsymbol{\Lambda}, \boldsymbol{\Psi}) = \log|\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}| + \operatorname{tr}\{\mathbf{S}_0(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1}\} + c.$$

For identifiability impose that $\boldsymbol{\Delta} = \boldsymbol{\Lambda}^\top \boldsymbol{\Psi}^{-1} \boldsymbol{\Lambda}$ is diagonal (if $\hat{\boldsymbol{\Lambda}}$ is an MLE, so is $\hat{\boldsymbol{\Lambda}}\mathbf{R}$).

Theorem 9.4: ($*$) $\boldsymbol{\Lambda}$ given $\boldsymbol{\Psi}$

Given $\boldsymbol{\Psi} = \operatorname{diag}(\psi_1, \dots, \psi_p)$, let $\mathbf{A} = \boldsymbol{\Psi}^{-1/2} \mathbf{S}_0 \boldsymbol{\Psi}^{-1/2} = \sum_{i=1}^{p} d_i \mathbf{e}_i \mathbf{e}_i^\top$ ($d_1 \geq \cdots \geq d_p$) and $\delta_i = d_i - 1 > 0$ ($i \leq k$). Then

$$\hat{\boldsymbol{\Lambda}} = \boldsymbol{\Psi}^{1/2} [\mathbf{e}_1, \dots, \mathbf{e}_k] \operatorname{diag}(\delta_1^{1/2}, \dots, \delta_k^{1/2}).$$

Proof [Supplement]. Let $\boldsymbol{\Lambda}^* = \boldsymbol{\Psi}^{-1/2}\boldsymbol{\Lambda}$ and $\boldsymbol{\Sigma}^* = \boldsymbol{\Psi}^{-1/2}\boldsymbol{\Sigma}\boldsymbol{\Psi}^{-1/2} = \boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top} + \mathbf{I}_p$. Then $\ell = \log|\boldsymbol{\Psi}| + \log|\boldsymbol{\Sigma}^*| + \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}^{*-1})$. If $\boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top}$ has eigenvalues $\gamma_1 \geq \cdots \geq \gamma_k \geq 0 = \cdots = 0$, $\boldsymbol{\Sigma}^*$ has eigenvalues $1 + \gamma_j$ and 1. Von Neumann's lower bound (pair the small eigenvalues of $\boldsymbol{\Sigma}^{*-1}$ with the large ones of $\mathbf{A}$) gives

$$\log|\boldsymbol{\Sigma}^*| + \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}^{*-1}) \geq \sum_{j=1}^{k} \Big\{ \log(1+\gamma_j) + \frac{d_j}{1+\gamma_j} \Big\} + \sum_{j>k} d_j,$$

with equality when the eigenvectors of $\boldsymbol{\Sigma}^*$ coincide with the $\mathbf{e}_j$ in the same order. Each term $\log x + d/x = \log d + \{d/x - \log(d/x)\} \geq \log d + 1$ (Theorem 2.13) is minimized at $x = d$, so $1 + \gamma_j = d_j$, i.e. $\gamma_j = \delta_j$. Hence $\boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top} = \sum_{j \leq k} \delta_j \mathbf{e}_j \mathbf{e}_j^\top$ and $\boldsymbol{\Lambda}^* = [\mathbf{e}_1, \dots, \mathbf{e}_k] \operatorname{diag}(\delta_j^{1/2})$ up to rotation; this choice makes $\boldsymbol{\Lambda}^{*\top}\boldsymbol{\Lambda}^* = \boldsymbol{\Delta}$ diagonal. ■

Theorem 9.5: ($**$) $\boldsymbol{\Psi}$ given $\boldsymbol{\Lambda}$

The MLE satisfies $\operatorname{diag}(\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}}) = \operatorname{diag}(\mathbf{S}_0)$ (the model reproduces the sample variances). Accordingly, given $\boldsymbol{\Lambda}$, update

$$\hat{\boldsymbol{\Psi}} = \operatorname{diag}(\hat{\psi}_1, \dots, \hat{\psi}_p), \qquad \hat{\psi}_i = (\mathbf{S}_0 - \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top)_{ii}.$$

Alternate the two steps. If some $\hat{\psi}_i \leq 0$ (a Heywood case), bound it below by a small positive number.

Logic for $(*)$ and $(**)$: the likelihood equations

We evaluate $\dfrac{\partial \ell(\mathbf{\Lambda},\mathbf{\Psi})}{\partial \mathbf{\Psi}} = 0$ and $\dfrac{\partial \ell(\mathbf{\Lambda},\mathbf{\Psi})}{\partial \mathbf{\Lambda}} = 0$ for

$$\ell(\mathbf{\Lambda},\mathbf{\Psi}) = \log|\mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi}| + \operatorname{tr}\{\mathbf{S}_0(\mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi})^{-1}\}, \qquad \mathbf{\Sigma} = \mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi},$$

using the facts that for a matrix $\mathbf{A} = \mathbf{A}(x)$,

$$\frac{\partial \mathbf{A}^{-1}}{\partial x} = -\mathbf{A}^{-1}\frac{\partial \mathbf{A}}{\partial x}\mathbf{A}^{-1}, \qquad \frac{\partial \log|\mathbf{A}|}{\partial x} = \operatorname{tr}\Big(\mathbf{A}^{-1}\frac{\partial \mathbf{A}}{\partial x}\Big)$$

(the second is the scalar form of Theorem 2.19; the first follows from differentiating $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$).

Derivative in ψ_i. With $\mathbf{e}_i$ the ith unit vector, $\frac{\partial}{\partial \psi_i}\operatorname{diag}(\psi_1,\dots,\psi_p) = \mathbf{e}_i\mathbf{e}_i^\top$, so $\frac{\partial \mathbf{\Sigma}}{\partial \psi_i} = \mathbf{e}_i\mathbf{e}_i^\top$ and

$$\begin{aligned}\frac{\partial \ell}{\partial \psi_i} &= \underbrace{\operatorname{tr}\Big(\mathbf{\Sigma}^{-1}\frac{\partial \mathbf{\Sigma}}{\partial \psi_i}\Big)}_{\partial \log|\mathbf{\Sigma}|/\partial\psi_i} + \underbrace{\operatorname{tr}\Big(\mathbf{S}_0\frac{\partial \mathbf{\Sigma}^{-1}}{\partial \psi_i}\Big)}_{\partial \operatorname{tr}(\mathbf{S}_0\mathbf{\Sigma}^{-1})/\partial\psi_i} = \operatorname{tr}(\mathbf{\Sigma}^{-1}\mathbf{e}_i\mathbf{e}_i^\top) - \operatorname{tr}(\mathbf{S}_0\mathbf{\Sigma}^{-1}\mathbf{e}_i\mathbf{e}_i^\top\mathbf{\Sigma}^{-1}) \\ &= \mathbf{e}_i^\top\mathbf{\Sigma}^{-1}\mathbf{e}_i - \mathbf{e}_i^\top\mathbf{\Sigma}^{-1}\mathbf{S}_0\mathbf{\Sigma}^{-1}\mathbf{e}_i = \mathbf{e}_i^\top(\mathbf{\Sigma}^{-1} - \mathbf{\Sigma}^{-1}\mathbf{S}_0\mathbf{\Sigma}^{-1})\mathbf{e}_i = 0, \qquad i = 1,\dots,p.\end{aligned}$$

$$\textcircled{1} \qquad \operatorname{diag}(\mathbf{\Sigma}^{-1} - \mathbf{\Sigma}^{-1}\mathbf{S}_0\mathbf{\Sigma}^{-1}) = \mathbf{0}.$$

Derivative in λ_{ij}. With $\mathbf{\Lambda} = [\mathbf{l}_1,\dots,\mathbf{l}_k]$ (columns), $\dfrac{\partial \mathbf{\Lambda}\mathbf{\Lambda}^\top}{\partial \lambda_{ij}} = \mathbf{l}_j\mathbf{e}_i^\top + \mathbf{e}_i\mathbf{l}_j^\top$. Then

$$\begin{aligned}\frac{\partial \ell}{\partial \lambda_{ij}} &= \operatorname{tr}\Big(\mathbf{\Sigma}^{-1}\frac{\partial \mathbf{\Lambda}\mathbf{\Lambda}^\top}{\partial \lambda_{ij}}\Big) - \operatorname{tr}\Big(\mathbf{S}_0\mathbf{\Sigma}^{-1}\frac{\partial \mathbf{\Sigma}}{\partial \lambda_{ij}}\mathbf{\Sigma}^{-1}\Big) \\ &= \operatorname{tr}(\mathbf{\Sigma}^{-1}\mathbf{l}_j\mathbf{e}_i^\top) + \operatorname{tr}(\mathbf{\Sigma}^{-1}\mathbf{e}_i\mathbf{l}_j^\top) - \operatorname{tr}(\mathbf{S}_0\mathbf{\Sigma}^{-1}\mathbf{l}_j\mathbf{e}_i^\top\mathbf{\Sigma}^{-1}) - \operatorname{tr}(\mathbf{S}_0\mathbf{\Sigma}^{-1}\mathbf{e}_i\mathbf{l}_j^\top\mathbf{\Sigma}^{-1}) \\ &= 2\mathbf{e}_i^\top\mathbf{\Sigma}^{-1}\mathbf{l}_j - 2\mathbf{e}_i^\top\mathbf{\Sigma}^{-1}\mathbf{S}_0\mathbf{\Sigma}^{-1}\mathbf{l}_j = 0 \qquad (i = 1,\dots,p,\ j = 1,\dots,k),\end{aligned}$$

using symmetry of $\mathbf{\Sigma}^{-1}$ and $\mathbf{\Sigma}^{-1}\mathbf{S}_0\mathbf{\Sigma}^{-1}$ (each pair of traces is equal). Collecting over i, j:

$$\mathbf{\Sigma}^{-1}\mathbf{\Lambda} = \mathbf{\Sigma}^{-1}\mathbf{S}_0\mathbf{\Sigma}^{-1}\mathbf{\Lambda} \overset{\text{premultiply } \mathbf{\Sigma}}{\Longrightarrow} \mathbf{\Lambda} = \mathbf{S}_0\mathbf{\Sigma}^{-1}\mathbf{\Lambda} = \mathbf{S}_0(\mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi})^{-1}\mathbf{\Lambda} = \mathbf{S}_0\mathbf{\Psi}^{-1}\mathbf{\Lambda}(\mathbf{I} + \mathbf{\Lambda}^\top\mathbf{\Psi}^{-1}\mathbf{\Lambda})^{-1},$$

where the last step uses $\mathbf{\Lambda}(\mathbf{I} + \mathbf{\Lambda}^\top\mathbf{\Psi}^{-1}\mathbf{\Lambda}) = (\mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi})\mathbf{\Psi}^{-1}\mathbf{\Lambda}$, i.e. $(\mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi})^{-1}\mathbf{\Lambda} = \mathbf{\Psi}^{-1}\mathbf{\Lambda}(\mathbf{I} + \mathbf{\Lambda}^\top\mathbf{\Psi}^{-1}\mathbf{\Lambda})^{-1}$.

Write $\mathbf{\Delta} \equiv \mathbf{\Lambda}^\top\mathbf{\Psi}^{-1}\mathbf{\Lambda}$ (and require $\mathbf{\Delta}_{k\times k}$ to be diagonal). Then

$$\textcircled{2}' \qquad \mathbf{\Lambda}(\mathbf{I} + \mathbf{\Delta}) = \mathbf{S}_0\mathbf{\Psi}^{-1}\mathbf{\Lambda}.$$

Premultiplying by $\mathbf{\Psi}^{-1/2}$:

$$\textcircled{2} \qquad \underbrace{\mathbf{\Psi}^{-1/2}\mathbf{S}_0\mathbf{\Psi}^{-1/2}}_{=:\mathbf{A}}\ \underbrace{\mathbf{\Psi}^{-1/2}\mathbf{\Lambda}}_{=:\tilde{\mathbf{E}}} = \underbrace{\mathbf{\Psi}^{-1/2}\mathbf{\Lambda}}_{\tilde{\mathbf{E}}}\ \underbrace{(\mathbf{I} + \mathbf{\Delta})}_{=:\mathbf{D}=\operatorname{diag}(d_1,\dots,d_k)}, \qquad \tilde{\mathbf{E}} = [\tilde{\mathbf{e}}_1,\dots,\tilde{\mathbf{e}}_k].$$

Hence $\mathbf{A}\tilde{\mathbf{e}}_i = \tilde{\mathbf{e}}_i d_i$, $i = 1,\dots,k$: $(\tilde{\mathbf{e}}_i, d_i)$ is an eigenpair of $\mathbf{A}$, although $\tilde{\mathbf{e}}_i^\top\tilde{\mathbf{e}}_i \neq 1$ in general. Since $\tilde{\mathbf{e}}_i^\top\tilde{\mathbf{e}}_i = (\tilde{\mathbf{E}}^\top\tilde{\mathbf{E}})_{ii} = (\mathbf{\Lambda}^\top\mathbf{\Psi}^{-1}\mathbf{\Lambda})_{ii} = \Delta_{ii}$, let $\mathbf{e}_i \equiv \tilde{\mathbf{e}}_i/\Delta_{ii}^{1/2}$ so that $\mathbf{e}_i^\top\mathbf{e}_i = 1$ and, for $\mathbf{E} = [\mathbf{e}_1,\dots,\mathbf{e}_k]$, $\mathbf{E}^\top\mathbf{E} = \mathbf{I}_k$ ($\mathbf{\Delta}$ diagonal). Then $(\mathbf{e}_i, d_i)$ is a unit eigenvector–eigenvalue pair of $\mathbf{A}$, and

$$\mathbf{D} = \mathbf{I} + \mathbf{\Delta}, \qquad \tilde{\mathbf{E}} = \mathbf{\Psi}^{-1/2}\mathbf{\Lambda} = \mathbf{E}\mathbf{\Delta}^{1/2}.$$

$(*)$ $\mathbf{\Psi}$ is fixed. From ②, $\mathbf{A} = \mathbf{\Psi}^{-1/2}\mathbf{S}_0\mathbf{\Psi}^{-1/2}$ is computable, with eigenpairs $(\mathbf{e}_i(\mathbf{A}), d_i(\mathbf{A}))$. Use $\mathbf{D} = \mathbf{I} + \mathbf{\Delta}$ to set $\hat{\mathbf{\Delta}} = \operatorname{diag}(d_i(\mathbf{A})) - \mathbf{I}$, and $\mathbf{E}\mathbf{\Delta}^{1/2} = \mathbf{\Psi}^{-1/2}\mathbf{\Lambda}$ to set

$$\hat{\mathbf{\Lambda}} = \mathbf{\Psi}^{1/2}\mathbf{E}(\mathbf{A})\hat{\mathbf{\Delta}}^{1/2}, \qquad \mathbf{E}(\mathbf{A}) = [\mathbf{e}_1(\mathbf{A}),\dots,\mathbf{e}_k(\mathbf{A})],$$

taking the k largest eigenvalues (the von Neumann argument above shows these give the minimum).

$(**)$ **$\boldsymbol{\Lambda}$ is fixed.** From ①, $\operatorname{diag}(\boldsymbol{\Sigma}^{-1} - \boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1}) = \mathbf{0}$.

Claim: if ②′ holds, then

$$\boldsymbol{\Sigma}^{-1} - \boldsymbol{\Sigma}^{-1}\mathbf{S}_0\boldsymbol{\Sigma}^{-1} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Sigma}^{-1} = \boldsymbol{\Psi}^{-1}(\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Psi}^{-1} = \boldsymbol{\Psi}^{-1}(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top - \mathbf{S}_0 + \boldsymbol{\Psi})\boldsymbol{\Psi}^{-1}.$$

Proof of claim [Supplement]. $\boldsymbol{\Sigma}\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda} = (\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda} = \boldsymbol{\Lambda}(\mathbf{I} + \boldsymbol{\Delta})$, and by ②′ also $\mathbf{S}_0\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda} = \boldsymbol{\Lambda}(\mathbf{I} + \boldsymbol{\Delta})$, so $(\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda} = \mathbf{0}$. By Woodbury, $\boldsymbol{\Sigma}^{-1} = \boldsymbol{\Psi}^{-1} - \boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}(\mathbf{I} + \boldsymbol{\Delta})^{-1}\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1}$, hence

$$(\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Sigma}^{-1} = (\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Psi}^{-1} - \underbrace{(\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}}_{=\mathbf{0}}(\mathbf{I} + \boldsymbol{\Delta})^{-1}\boldsymbol{\Lambda}^\top\boldsymbol{\Psi}^{-1} = (\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Psi}^{-1},$$

and transposing, $\boldsymbol{\Sigma}^{-1}(\boldsymbol{\Sigma} - \mathbf{S}_0) = \boldsymbol{\Psi}^{-1}(\boldsymbol{\Sigma} - \mathbf{S}_0)$. Therefore $\boldsymbol{\Sigma}^{-1}(\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Sigma}^{-1} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Psi}^{-1} = \boldsymbol{\Psi}^{-1}(\boldsymbol{\Sigma} - \mathbf{S}_0)\boldsymbol{\Psi}^{-1}$.

Since $\boldsymbol{\Psi}^{-1}$ is diagonal, the ith diagonal entry of $\boldsymbol{\Psi}^{-1}(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top - \mathbf{S}_0 + \boldsymbol{\Psi})\boldsymbol{\Psi}^{-1}$ is $(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top - \mathbf{S}_0 + \boldsymbol{\Psi})_{ii}/\psi_i^2$, so ① becomes

$$\operatorname{diag}(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top - \mathbf{S}_0 + \boldsymbol{\Psi}) = \mathbf{0} \quad \Longrightarrow \quad \psi_i = (\mathbf{S}_0 - \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top)_{ii},$$

which is the update $(**)$ and also shows $\operatorname{diag}(\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}}) = \operatorname{diag}(\mathbf{S}_0)$ at the MLE.

2.4 Likelihood ratio test for the number of factors

$H_0 : \boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}$ (k factors) versus H_1: unrestricted $\boldsymbol{\Sigma}$. With $L_1 = L(\bar{\mathbf{x}}, \mathbf{S}_0)$ and $L_0 = L(\bar{\mathbf{x}}, \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}})$,

$$W = -2\log\frac{L_0}{L_1} = n\log\Big(\frac{|\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}}|}{|\mathbf{S}_0|}\Big) \;\dot{\sim}\; \chi^2_\nu,$$

$$\nu = \frac{p(p+1)}{2} - \Big\{pk + p - \frac{k(k-1)}{2}\Big\} = \frac{(p-k)^2 - p - k}{2}.$$

(At the MLE $\operatorname{tr}\{\mathbf{S}_0\hat{\boldsymbol{\Sigma}}^{-1}\} = p$, so the trace terms cancel.) With Bartlett's correction, reject the k-factor model if $\big(n - 1 - \frac{2p+4k+5}{6}\big)\ln\frac{|\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}}|}{|\mathbf{S}_0|} > \chi^2_\nu(\alpha)$.

3 Factor Rotation

Non-identifiability is used to find interpretable loadings, ideally a simple structure in which each variable loads heavily on only one factor. Rotate $\boldsymbol{\Lambda}^* = \boldsymbol{\Lambda}\mathbf{J}$ with $\mathbf{J}$ orthogonal.

Definition 9.6: varimax and quartimax

Let $\tilde{\lambda}^*_{i\ell} = \lambda^*_{i\ell}/h_i$ (normalized by the communality).

$$\text{Varimax: } \mathbf{J}^\dagger = \arg\max_{\mathbf{J}} \sum_{\ell=1}^{k} \widehat{\operatorname{Var}}\big((\tilde{\lambda}^*_{1\ell})^2, \ldots, (\tilde{\lambda}^*_{p\ell})^2\big)$$

— spread the squared loadings within each *column* (factor), so each factor has some large and some near-zero loadings.

$$\text{Quartimax: } \mathbf{J}^\dagger = \arg\max_{\mathbf{J}} \sum_{i=1}^{p} \widehat{\operatorname{Var}}\big((\lambda^*_{i1})^2, \ldots, (\lambda^*_{ik})^2\big)$$

— spread the squared loadings within each *row* (variable), so each variable concentrates on one factor.

Rotation leaves the communalities h_i^2 and $\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top$ unchanged.

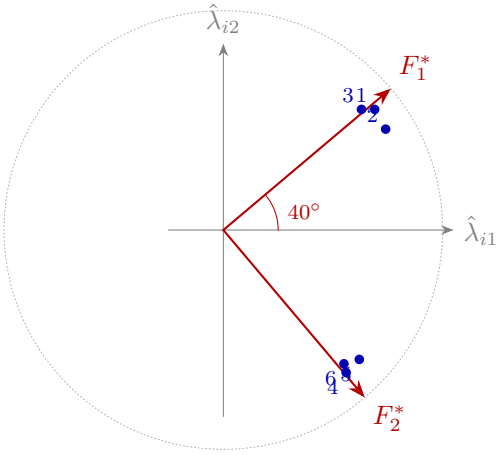


Figure. *Loadings of the six variables of the previous figure on the unrotated principal component factors (blue). Every variable loads about* 0.6 *on* F_1*, so* F_1 *looks like a general factor. Varimax rotates the axes by* 40° *(red); in the new coordinates the loadings are about* $(0.86, 0)$ *for variables 1–3 and* $(0, 0.85)$ *for variables 4–6, a simple structure. The dotted unit circle bounds the communalities, which rotation does not change.*

4 Factor Score Prediction

4.1 Weighted least squares (Bartlett) scores

(1) Choose $\hat{\mathbf{F}}$ to minimize the distance between $\mathbf{X}$ and $\boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{F}$: with the estimates fixed, regard $\mathbf{x}_i - \bar{\mathbf{x}} = \hat{\boldsymbol{\Lambda}}\mathbf{F}_i + \mathbf{U}_i$ as a heteroscedastic regression,

$$\hat{\mathbf{F}}_i^{\text{wls}} = \arg\min_{\mathbf{F}_i} (\mathbf{x}_i - \bar{\mathbf{x}} - \hat{\boldsymbol{\Lambda}}\mathbf{F}_i)^\top \hat{\boldsymbol{\Psi}}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}} - \hat{\boldsymbol{\Lambda}}\mathbf{F}_i) = (\hat{\boldsymbol{\Lambda}}^\top \hat{\boldsymbol{\Psi}}^{-1} \hat{\boldsymbol{\Lambda}})^{-1} \hat{\boldsymbol{\Lambda}}^\top \hat{\boldsymbol{\Psi}}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}})$$

(set the gradient of the quadratic form to zero).

4.2 Regression scores

(2) Since $\begin{pmatrix} \mathbf{X} \\ \mathbf{F} \end{pmatrix} \sim N_{p+k}\Big(\begin{pmatrix} \boldsymbol{\mu} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi} & \boldsymbol{\Lambda} \\ \boldsymbol{\Lambda}^\top & \mathbf{I}_k \end{pmatrix} \Big)$, the conditional expectation (Theorem 3.12) gives

$$\hat{\mathbf{F}}_i^{\text{reg}} = \mathrm{E}(\mathbf{F} \mid \mathbf{X} = \mathbf{x}_i) = \hat{\boldsymbol{\Lambda}}^\top (\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}})^{-1} (\mathbf{x}_i - \bar{\mathbf{x}}).$$

Theorem 9.7: relation between the two scores

With $\boldsymbol{\Delta} = \boldsymbol{\Lambda}^\top \boldsymbol{\Psi}^{-1} \boldsymbol{\Lambda}$, $\hat{\mathbf{F}}^{\text{reg}} = (\mathbf{I} + \boldsymbol{\Delta})^{-1} \boldsymbol{\Delta}\, \hat{\mathbf{F}}^{\text{wls}} = (\mathbf{I} + \boldsymbol{\Delta}^{-1})^{-1} \hat{\mathbf{F}}^{\text{wls}}$. If $\boldsymbol{\Delta}$ is diagonal each component is shrunk by $\delta_\ell/(1+\delta_\ell) < 1$, so $|\hat{F}_\ell^{\text{wls}}| \geq |\hat{F}_\ell^{\text{reg}}|$.

Proof [Supplement]. By Woodbury, $(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1} = \boldsymbol{\Psi}^{-1} - \boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}(\mathbf{I} + \boldsymbol{\Delta})^{-1}\boldsymbol{\Lambda}^\top \boldsymbol{\Psi}^{-1}$, so

$$\boldsymbol{\Lambda}^\top (\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi})^{-1} = \{\mathbf{I} - \boldsymbol{\Delta}(\mathbf{I} + \boldsymbol{\Delta})^{-1}\} \boldsymbol{\Lambda}^\top \boldsymbol{\Psi}^{-1} = (\mathbf{I} + \boldsymbol{\Delta})^{-1} \boldsymbol{\Lambda}^\top \boldsymbol{\Psi}^{-1}$$

(as $\mathbf{I} - \boldsymbol{\Delta}(\mathbf{I} + \boldsymbol{\Delta})^{-1} = (\mathbf{I} + \boldsymbol{\Delta} - \boldsymbol{\Delta})(\mathbf{I} + \boldsymbol{\Delta})^{-1}$), while $\boldsymbol{\Lambda}^\top \boldsymbol{\Psi}^{-1}(\mathbf{x} - \bar{\mathbf{x}}) = \boldsymbol{\Delta}\hat{\mathbf{F}}^{\text{wls}}$. ■

Regression scores shrink towards 0, reflecting the prior $\mathbf{F} \sim N(\mathbf{0}, \mathbf{I})$ (a Bayes estimate); WLS scores treat $\mathbf{F}$ as a fixed parameter and are conditionally unbiased.

5 PCA versus Factor Analysis

[Supplement] The table collects the differences met in Ch. 7 and in this chapter.

	PCA	Factor analysis
Goal	linear combinations explaining total variance $\mathrm{tr}\boldsymbol{\Sigma}$	latent factors explaining covariances (off-diagonal)
Model	none (descriptive transformation)	$\boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}$
Direction	variables $\to$ components ($\mathbf{Z} = \mathbf{U}^\top\mathbf{X}$)	factors $\to$ variables ($\mathbf{X} = \boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{F} + \mathbf{U}$)
Uniqueness	unique if eigenvalues distinct	only up to rotation
Increasing k	earlier components unchanged	(MLE) earlier loadings change

6 Selected exercises

Motivation. The exercises prove two claims made in passing above, the residual bound of the principal component method and the simplification of the likelihood ratio statistic, and show by the smallest example why a factor model need not be identified.

Exercise 9.1 (Residuals of the principal component method; J&W 9.5)

Let $\mathbf{R} = \sum_{j=1}^{p} l_j \mathbf{u}_j \mathbf{u}_j^\top$ $(l_1 \geq \cdots \geq l_p)$, $\hat{\boldsymbol{\Lambda}} = [\sqrt{l_1}\mathbf{u}_1, \ldots, \sqrt{l_k}\mathbf{u}_k]$, $\hat{\boldsymbol{\Psi}} = \mathrm{diag}(\mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top)$ and $\mathbf{E} = \mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top - \hat{\boldsymbol{\Psi}}$. Show

$$\sum_{i,j} E_{ij}^2 \leq l_{k+1}^2 + \cdots + l_p^2.$$

Solution.

Step 1 (zero diagonal). By the definition of $\hat{\boldsymbol{\Psi}}$, $E_{ii} = 0$, and $E_{ij} = (\mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top)_{ij}$ for $i \neq j$. Hence

$$\|\mathbf{E}\|_F^2 = \sum_{i \neq j} (\mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top)_{ij}^2 \leq \|\mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top\|_F^2.$$

Step 2 (the discarded part). $\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top = \sum_{j \leq k} l_j \mathbf{u}_j \mathbf{u}_j^\top$, so $\mathbf{R} - \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top = \mathbf{P}_2 \mathbf{L}_2 \mathbf{P}_2^\top$ with $\mathbf{P}_2 = [\mathbf{u}_{k+1}, \ldots, \mathbf{u}_p]$, $\mathbf{P}_2^\top \mathbf{P}_2 = \mathbf{I}$ and $\mathbf{L}_2 = \mathrm{diag}(l_{k+1}, \ldots, l_p)$.

Step 3. By Theorem 2.3(ii)–(iii), $\|\mathbf{P}_2 \mathbf{L}_2 \mathbf{P}_2^\top\|_F^2 = \mathrm{tr}(\mathbf{P}_2 \mathbf{L}_2 \mathbf{P}_2^\top \mathbf{P}_2 \mathbf{L}_2 \mathbf{P}_2^\top) = \mathrm{tr}(\mathbf{L}_2^2 \mathbf{P}_2^\top \mathbf{P}_2) = \sum_{j>k} l_j^2$.

The same argument applies with $\mathbf{S}$ in place of $\mathbf{R}$. ■

Exercise 9.2 (Non-uniqueness for $p = 2$, $k = 1$; J&W 9.7)

For $p = 2$ and $k = 1$ the model reads $\sigma_{11} = \lambda_1^2 + \psi_1$, $\sigma_{12} = \lambda_1\lambda_2$, $\sigma_{22} = \lambda_2^2 + \psi_2$. Given $\boldsymbol{\Sigma} \succ 0$ with $\sigma_{12} \neq 0$, find all solutions with $\psi_1, \psi_2 \geq 0$ and show that there are infinitely many. Relate this to the sign of ν.

Solution.

Step 1. $\sigma_{12} \neq 0$ forces $\lambda_1 = t \neq 0$ and $\lambda_2 = \sigma_{12}/t$; then $\psi_1 = \sigma_{11} - t^2$ and $\psi_2 = \sigma_{22} - \sigma_{12}^2/t^2$ are determined.

Step 2. $\psi_1 \geq 0$ and $\psi_2 \geq 0$ iff $\sigma_{12}^2/\sigma_{22} \leq t^2 \leq \sigma_{11}$. Since $\boldsymbol{\Sigma} \succ 0$, $\sigma_{12}^2 < \sigma_{11}\sigma_{22}$, so this is a non-degenerate interval for t^2, and every t in it (with either sign) gives a valid $(\boldsymbol{\Lambda}, \boldsymbol{\Psi})$.

Step 3. For $k = 1$ the only orthogonal $\mathbf{T}$ is ± 1, so rotation explains just the sign of t; the continuum of solutions is a genuine lack of identification. This matches $\nu = \{(p-k)^2 - p - k\}/2 = (1-3)/2 = -1 < 0$: three equations for four parameters. ■

Exercise 9.3 (Simplifying the likelihood ratio statistic)

Let $(\hat{\boldsymbol{\Lambda}}, \hat{\boldsymbol{\Psi}})$, $\hat{\boldsymbol{\Psi}} \succ 0$, solve the likelihood equations ① and ②′, and $\hat{\boldsymbol{\Sigma}} = \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}}$. Show $\operatorname{tr}(\mathbf{S}_0\hat{\boldsymbol{\Sigma}}^{-1}) = p$, and deduce $W = -2\log(L_0/L_1) = n\log(|\hat{\boldsymbol{\Sigma}}|/|\mathbf{S}_0|)$.

Solution [Supplement].

Step 1. The proof of the Claim above shows that ②′ implies $\hat{\boldsymbol{\Sigma}}^{-1}(\hat{\boldsymbol{\Sigma}} - \mathbf{S}_0) = \hat{\boldsymbol{\Psi}}^{-1}(\hat{\boldsymbol{\Sigma}} - \mathbf{S}_0)$, and ① then gives $\operatorname{diag}(\hat{\boldsymbol{\Sigma}}) = \operatorname{diag}(\mathbf{S}_0)$.

Step 2. Since $\hat{\boldsymbol{\Psi}}^{-1}$ is diagonal,

$$p - \operatorname{tr}(\hat{\boldsymbol{\Sigma}}^{-1}\mathbf{S}_0) = \operatorname{tr}\{\hat{\boldsymbol{\Sigma}}^{-1}(\hat{\boldsymbol{\Sigma}} - \mathbf{S}_0)\} = \operatorname{tr}\{\hat{\boldsymbol{\Psi}}^{-1}(\hat{\boldsymbol{\Sigma}} - \mathbf{S}_0)\} = \sum_{i=1}^{p} \frac{(\hat{\boldsymbol{\Sigma}} - \mathbf{S}_0)_{ii}}{\hat{\psi}_i} = 0.$$

Step 3. With $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$, $L(\bar{\mathbf{x}}, \boldsymbol{\Sigma}) = (2\pi)^{-np/2}|\boldsymbol{\Sigma}|^{-n/2}\exp\{-\frac{n}{2}\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0)\}$, so

$$-2\log\frac{L_0}{L_1} = n\log\frac{|\hat{\boldsymbol{\Sigma}}|}{|\mathbf{S}_0|} + n\{\operatorname{tr}(\hat{\boldsymbol{\Sigma}}^{-1}\mathbf{S}_0) - \operatorname{tr}(\mathbf{S}_0^{-1}\mathbf{S}_0)\} = n\log\frac{|\hat{\boldsymbol{\Sigma}}|}{|\mathbf{S}_0|},$$

because both traces equal p. As $L_1 \geq L_0$, also $|\hat{\boldsymbol{\Sigma}}| \geq |\mathbf{S}_0|$. ■